Miaoyi Liao

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Research Interests: Medical AI, LLM Agents, Trustworthy Machine Learning, Knowledge Graphs

Education

Southern University of Science and Technology (SUSTech) 2023 – Present

B.S. in *Data Science and Big Data Technology*, expected March 2028
Previous training in Clinical Medicine, 2023–2025
GPA: 3.90 / 4.00 (Top 10%)

Research Experience

Department of Statistics and Data Science, SUSTech **Feb 2026 – Present**

Undergraduate Researcher *Supervised by Prof. Guanhua Chen*

Medical AI Agents

- Converted **PrimeKG** from CSV files into a **Neo4j** knowledge graph to support structured medical knowledge retrieval and Cypher-based querying.
- Reproduced a **knowledge-graph-grounded agent pipeline** (*User Question → CLI → ReAct Agent → Neo4j MCP Server → Neo4j/Cypher → Results*) using a DashScope model, enabling natural-language medical queries over the graph.
- Previously reproduced the inference workflow of an **Alibaba WebAgent** and built an **AfriMedQA-based evaluation pipeline** as a foundation for medical agent experimentation.

Department of Computer Science, SUSTech **Sep 2025 – Feb 2026**

Undergraduate Researcher *Supervised by Prof. Jianguo Zhang*

Out-of-Distribution (OOD) Detection & Trustworthy Machine Learning

- Developed a perturbation-driven OOD detection method in OpenOOD using gradient-based semantic/covariate perturbations at BatchNorm layers with two auxiliary heads, improving AUROC by up to 5.3 points and reducing FPR@95 by up to 18.6 points over the MSP baseline on CIFAR-10 benchmarks.
- Ran ablation studies on gradient processing, perturbation layer placement, and backpropagation scope; identified gradient clearing as harmful and validated auxiliary-head initialization, improving AUROC from 91.84 to 92.58 and lowering FPR@95 from 26.29 to 24.14.

Projects

Qwen3-Math RL Pipeline

LLM Course Mid-term Project | GitHub

- Built an end-to-end optimization pipeline for Qwen3-Math, covering math data cleaning, prompt-completion formatting, LoRA-based supervised fine-tuning, inference evaluation, and RL training preparation.
- Implemented local multi-pass evaluation to compute `pass@1`, `pass@8`, and `correct_ratio`, then stratified samples by difficulty to identify unstable reasoning cases suitable for reinforcement learning.
- Prepared RL training data and implemented GRPO training scripts, improving final validation accuracy from 38.5% to 46.70% (+8.20 percentage points).

Evidence Chain Verification System for Research Agents

LLM Course Final Project | DeepScientist Extension

- Extended DeepScientist with a citation fact-checking pipeline for research-agent reports, including PDF claim extraction, evidence tracking, literature API verification, and auditable report generation.
- Implemented a Semantic Scholar → arXiv → Crossref fallback verifier and structured claim-verification workflow to trace claims back to external evidence.

- Added framework-level execution constraints to prevent LLM agents from bypassing mandatory scoring, rendering, artifact recording, and memory-writing steps.
- Validated the system with 139 automated tests and 6 real multi-domain quests, covering 144 claim-verification calls and achieving 36/36 mandatory tool-call compliance.

Optimization of NIPT Testing Timing and Fetal Chromosomal Abnormality Diagnosis
Mathematical Contest in Modeling Project | First Prize (Provincial Level)

- Modeled the relationship among gestational age, BMI, and fetal Y-chromosome concentration using a linear mixed-effects model with logit transformation, achieving $R^2 = 0.789$.
- Applied clustering, constrained optimization, and logistic regression to recommend BMI-specific NIPT testing windows and detect fetal chromosomal abnormalities, achieving AUC = 0.822.

Chinese Chess Game System
Course Project | Selected as one of two showcased projects in a class of 150+ students

- Developed a Chinese Chess application with account login, save/load, replay, and four gameplay modes: timed, human-vs-AI, hot-seat, and classic endgames.
- Designed the user interface and implemented core piece-rule logic, greedy algorithm-based AI gameplay, and replay functionality.

Honors and Awards

Outstanding Student Scholarship (Top 3%)	2023–2025
First Prize in National Undergraduate Mathematical Contest in Modeling (Provincial Level)	2025
Silver Medal, iGEM (International Genetically Engineered Machine Competition)	2025
First Prize in Undergraduate Basic Medical Innovation Competition (University Level)	2025
Outstanding Student Representative, SUSTech	May 2024
Outstanding Student Leader, SUSTech	May 2024

Skills

- Programming: Python, Java, SQL/Cypher
- Machine Learning / Data Analysis: pandas, NumPy, statsmodels
- Knowledge Graph / Database: Neo4j, Knowledge Graph Construction, Cypher Querying
- LLM / Agent Systems: ReAct prompting, DashScope API, Neo4j-grounded agent workflows, benchmark and evaluation pipelines
- Tools: Git, Linux, Docker, LaTeX
- Languages: TOEFL 108 | CET-6 620+ | English Placement Level A